

Worksheet: Reinforcement Learning & Q-Learning [SOLUTION]**Problem 1: Discounted Returns & Recursion**

Problem Setup: An agent operates in an episodic environment with a discount factor of $\gamma = 0.9$. Starting from time $t = 0$, it receives the following sequence of immediate rewards before the episode terminates at time step $T = 6$:

$$R_1 = +10, \quad R_2 = -5, \quad R_3 = +20, \quad R_4 = +5, \quad R_5 = -10, \quad R_6 = +15$$

Given Equations:

- Total Discounted Return: $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$
- Recursive Relationship: $G_t = R_{t+1} + \gamma G_{t+1}$

Questions: Calculate the exact total discounted return G_t for the following time steps. Show your work in the space provided and write your final answer on the line.

(a) $t = 5$

Calculation:

$$G_5 = R_6 = 15.0$$

Final Answer: $G_5 = \underline{15.0}$

(b) $t = 3$

Calculation:

$$\begin{aligned} G_3 &= R_4 + \gamma R_5 + \gamma^2 R_6 \\ &= 5 + (0.9)(-10) + (0.81)(15) \\ &= 5 - 9.0 + 12.15 = 8.15 \end{aligned}$$

Final Answer: $G_3 = \underline{8.15}$

(c) $t = 2$

Calculation:

$$\begin{aligned} G_2 &= R_3 + \gamma G_3 \quad (\text{using recursive relationship}) \\ &= 20 + 0.9(8.15) = 20 + 7.335 = 27.335 \end{aligned}$$

Final Answer: $G_2 = \underline{27.335}$

(d) $t = 0$

Calculation:

$$\begin{aligned} G_0 &= R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \gamma^4 R_5 + \gamma^5 R_6 \\ \text{or } G_0 &= R_1 + \gamma(R_2 + \gamma G_2) \\ G_1 &= -5 + 0.9(27.335) = -5 + 24.6015 = 19.6015 \\ G_0 &= 10 + 0.9(19.6015) = 27.6415 \end{aligned}$$

Final Answer: $G_0 = \underline{27.6415}$

Problem 2: Q-Learning (Off-Policy TD Update)

Problem Setup: An agent is training using tabular Q-learning (using a Q-table) with a discount factor $\gamma = 0.8$ and a learning rate $\alpha = 0.25$. The agent transitions from State A to State B by taking action "East", receiving a reward of +4.0. Transition: ($S = A$, $A = \text{East}$, $R = +4.0$, $S' = B$). The current Q-table has the following estimates:

$$Q(A, \text{East}) = 6.0, \quad Q(B, \text{North}) = 5.0, \quad Q(B, \text{South}) = 2.0, \quad Q(B, \text{West}) = 7.5$$

Given Equations (Q-Learning):

- TD Target $= R + \gamma \max_{a'} Q(S', a')$
- TD Error (δ) $= \text{TD Target} - Q(S, A)$
- Update Rule: $Q(S, A) \leftarrow Q(S, A) + \alpha \cdot \delta$

Questions: Calculate the Temporal-Difference (TD) Target for this transition. Next, calculate the TD Error (δ). Finally, calculate the new updated value for $Q(A, \text{East})$. Show your work in the space provided and write your final answer on the line.

(a) TD Target

Calculation:

$$\begin{aligned} \text{TD Target} &= R + \gamma \max_{a'} Q(S', a') \\ &= 4.0 + 0.8 \cdot \max(5.0, 2.0, 7.5) = 4.0 + 0.8 \cdot (7.5) = 4.0 + 6.0 = 10.0 \end{aligned}$$

Final Answer: TD Target = 10.0

(b) TD Error (δ)

Calculation:

$$\begin{aligned} \delta &= \text{TD Target} - Q(S, A) \\ &= 10.0 - 6.0 = 4.0 \end{aligned}$$

Final Answer: $\delta = \underline{4.0}$

(c) Q-Value Update

Calculation:

$$\begin{aligned} Q(A, \text{East}) &\leftarrow Q(A, \text{East}) + \alpha \cdot \delta \\ &= 6.0 + 0.25 \cdot (4.0) = 6.0 + 1.0 = 7.0 \end{aligned}$$

Final Answer: $Q(A, \text{East}) = \underline{7.0}$